

1. Executive Summary

RoamCraft is an intelligent web application designed to simplify vacation planning. Powered by **Google Gemini 2.5 Flash** and built on a high-performance **Node.js & Express** backend, RoamCraft allows users to generate comprehensive, day-by-day travel itineraries in real-time streaming mode.

-  **Live Deployment:** <https://roamcraft.onrender.com>
-  **GitHub Repository:** <https://github.com/sagun-glitch/RoamCraft>

2. Key Features

- Real-Time AI Streaming (Server-Sent Events):** Delivers instant markdown-formatted itinerary updates chunk-by-chunk without long loading waits.
- Customizable Parameters:** Flexible inputs for Destination, Trip Duration (1-14 days), Budget Tiers (Budget, Moderate, Luxury), and Personal Interests.
- Interactive Day-by-Day Dashboard:** Expandable timeline cards, morning/afternoon/evening schedule nodes, packing checklists, and local tips.
- One-Click Export & Print:** Styled PDF/Print document export and clipboard text copying.
- Scandinavian Terracotta Sand Theme:** High-contrast, clean UI designed for maximum readability without gradients or glassmorphism.

3. Technical Stack & Deployment

Component	Technology	Description
Live Deployment	Render Cloud Hosting	Deployed live at https://roamcraft.onrender.com
Backend Framework	Node.js / Express.js	Handles REST endpoints, static asset serving, and SSE streaming.
AI Model	Google Gemini 2.5 Flash	High-speed generative AI for structuring personalized itineraries.

Component	Technology	Description
Frontend UI	HTML5 / Vanilla CSS / ES6+ JS	Lightweight dashboard interface with Lucide Icons & Google Fonts.
Data Protocol	Server-Sent Events (SSE)	Enables real-time chunked streaming output to client browser.

4. API Specification

POST `/api/generate-itinerary`

Request Body (JSON):

```
{
  "destination": "Paris, France",
  "days": 3,
  "budget": "Moderate",
  "interests": "museums, bakeries, evening walks"
}
```

5. Setup & Local Installation

- Clone repository: `git clone https://github.com/sagun-glitch/RoamCraft.git`
- Install dependencies: `npm install`
- Add `.env` file with `GEMINI_API_KEY=YOUR_KEY`
- Start server: `npm start` (Open `http://localhost:3000` or visit `https://roamcraft.onrender.com`)